Miles Choquette

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EDUCATION

Northeastern University Boston, MA

Master of Science, Quantitative Finance | GPA 3.6 Expected Dec 2026

- Coursework: Fixed Income, Quantitative Portfolio Management, Empirical Methods in Finance, Advanced Topics in Finance

Florida Atlantic University Boca Raton, FL

Bachelor of Science, Computer Science; Certificate in Artificial Intelligence | GPA 3.45 May 2025

RESEARCH

Nelson-Siegel / Diebold-Li Treasury ETF Rotation | *Python, statsmodels* Spring 2026

- Fit Nelson-Siegel to monthly Treasury par curves (2008 to 2024) and forecast level, slope and curvature with an expanding-window VAR(1); combined curve-mispricing and directional signals to allocate across 7 Treasury ETFs.
- Top-3 portfolio earned a Sharpe ratio of 0.39 vs. 0.29 for equal weight, and 1.09 vs. 0.59 over 2018 to 2021; factor attribution showed slope carries the predictive power. Tested significance with Jobson-Korkie.

Polymarket Earnings Signal | *Python* 2026

- Built a no-look-ahead pipeline over 1,231 resolved earnings-beat contracts on ~415 tickers; ran calibration tests, an encompassing regression and an event study against a firm-specific beat-rate baseline.
- Market probabilities beat the baseline (Brier 0.149 vs. 0.185; coefficient 5.78, $p < 10^{-35}$), but permutation, block-bootstrap and Jobson-Korkie tests showed the trading edge was not significant once same-day clustering cut the sample to ~181 independent dates.

Greeks Across Credit Instruments | *Python* 2026

- Priced 7 credit instruments (fixed and callable bonds, FRN, revolver, CDS, mortgage, auto loan) in one survival-discounting framework with a bump-and-reprice engine for rate, spread, hazard, recovery, prepayment and gamma.
- Quantified embedded-option convexity: at −200bp a fixed bond gains 9.1 per 100 while a prepayable mortgage gains only 1.8.

Premarket Small-Cap Event Model | *Python, XGBoost* 2026

- Engineered 24 features from 1-second bars of small-cap premarket spikes; predicted rest-of-day moves with out-of-sample rank correlation of 0.205 ($p = 5\times10^{-8}$).
- Showed that buying premarket pumps is structurally negative expected value (median −9.4% open to close) and repurposed the model as a risk filter that flags the worst 40% of setups.

EXPERIENCE

Adicot Engineering | *Software Engineer & Junior Designer* May 2025 – Present

- Sole developer of the firm's Flask platform for live quote pricing, client intake, job tracking and signed PDF deliverables; designed pricing to fall back to known rates so a bad edit can never produce a wrong quote.
- Built load-calculation software that parses vector geometry from CAD-exported PDFs into rooms and runs ASHRAE Radiant Time Series cooling and heating loads, replacing work previously spread across three programs.
- At a three-person firm, learned new skills on the fly to cover software, HVAC design and client deliverables.

Draper Laboratory | *Student Undergraduate Engineer* May 2024 – May 2025

- Designed an inspection interface and the backend behind it, automating manual data entry and analysis, with potential to cut a 400-hour inspection process to about 40 hours.
- Built a database logging user interactions with the tool to support data-driven process decisions.

360 Huntington Fund | *Analyst, Portfolio & Research* Sep 2025 – 2026

- Pitched Buy recommendations on Waystar (WAY) and JFrog (FROG) to the investment committee; first pitch passed unanimously.
- Rebuilt the portfolio dashboard for the ~$2M, 46-position student-run fund, improving data accuracy and reporting.

SKILLS

Programming: Python (pandas, NumPy, SciPy, statsmodels, scikit-learn, XGBoost), SQL, MATLAB, Java, C++, Flask, Git
Quantitative methods: Yield-curve modeling, VAR, survival and hazard-rate pricing, backtesting, bootstrap and permutation inference, machine learning
Tools: Bloomberg Terminal, Jupyter, Excel